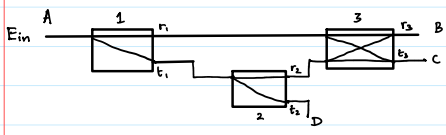
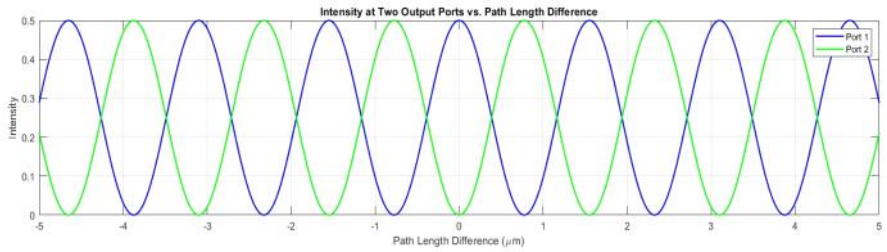
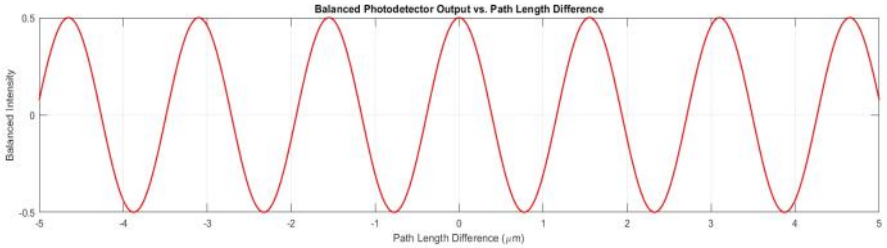
LDV interferometer

Monday, November 11, 2024 5:22 PM







1st Coupler: (50-50)

$$E_A = e^{i\phi_0} r e^{i\phi_0} + e^{i\phi_0} t e^{\phi_2} t' e^{i\phi_0}$$

$$E_D = e^{i\phi_0} r e^{i\phi_0} + e^{i\phi_0} t e^{i\phi_2} r' e^{i\phi_3}$$

$$I_A = |E_A|^2 \quad ; \quad I_D = |E_D|^2$$

$$r = |r| e^{i\theta_r} \quad ; \quad r' = |r'| e^{i\theta'_r}$$
$$t = |t| e^{i\theta_t} \quad ; \quad t' = |t'| e^{i\theta'_t}$$

Energy Conservation,

$$1 = |r|^2 (|r^2| + |t|^2) + |t|^2 (|t'|^2 + |r'|^2)$$
$$+ 2|r|^2 |tt'| \cos(\theta_t + \theta'_t - 2\theta_r + \phi_2 - \phi_1)$$
$$+ 2|rr't^2| \cos(\theta_r' - \theta_r + \phi_2 - \phi_1)$$

$$E_A = e^{i\phi_0} \left(|r| e^{i\theta_r} e^{i\phi_0} + |t| e^{i\theta_t} e^{i\phi_2} |t'| e^{i\theta'_t} e^{i\phi_0} \right)$$
$$= e^{i(2\phi_0)} \left(|r| e^{i\theta_r} + |tt'| e^{i(\theta_t + \theta'_t + \phi_2 - 2\phi_0)} \right)$$

$$E_D = e^{i\phi_0} \left(|r| e^{i\theta_r} e^{i\phi_0} + |t| e^{i\theta_t} e^{i\phi_2} |r'| e^{i\theta'_r} e^{i\phi_3} \right)$$
$$= e^{i(2\phi_0)} \left(|r| e^{i\theta_r} + |tr'| e^{i(\theta_t + \theta'_r + \phi_2 + \phi_3 - 2\phi_0)} \right)$$

$$I_A = |r|^2 + |t|^2 |t'|^2 + 2|r||tt'| \cos(\theta_t + \theta'_t - \theta_r + \phi_2 - 2\phi_0)$$

$$I_D = |r|^2 + |t|^2 |r'|^2 + 2|r||tr'| \cos(\theta'_r - \theta_r + \theta_t + \phi_2 + \phi_3 - 2\phi_0)$$

$$\Rightarrow |r|^2 (|r|^2 + |t|^2) + |t|^2 (|t'|^2 + |r'|^2) + 2|r||tt'| \cos(\theta_t + \theta'_t - \theta_r + \phi_2 - 2\phi_0)$$
$$+ 2|r||r't| \cos(\theta'_r - \theta_r + \phi_2 + \phi_3 - 2\phi_0)$$

$$= 1$$

- 90-10 Coupler

Simplify. $E_{D_{90\%}} = E_D \, t_{90}$

$$t_{90} = |t_{90}| e^{i\theta_{t_{90}}}$$

$$= E_D |t_{90}| e^{i\theta_{t_{90}}}$$

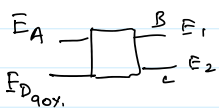
$$E_{D_{10\%}} = E_D \cdot r_{10}$$

$$|r_{10}| = |r_{10}| e^{i\theta_{r_{10}}} \qquad = E_D \cdot |r_{10}| e^{i\theta_{r_{10}}}$$

$$E_{D_{10\%}} = e^{i(2\phi_0)} \left(|r| e^{i\theta_r} + |tr'| e^{i(\theta_t + \theta'_r + \phi_2 + \phi_3 - 2\phi_0)} |r_{10}| e^{i\theta_{r_{10}}} \right)$$

$$E_{D_{90\%}} = e^{i(2\phi_0)} \left(|r| e^{i\theta_r} + |tr'| e^{i(\theta_t + \theta'_r + \phi_2 + \phi_3 - 2\phi_0)} |t_{90}| e^{i\theta_{t_{90}}} \right)$$

- Final Coupler (50-50)



$$E_1 = E_A \cdot t + E_{D_{90}} \cdot r$$

$$= e^{i(2\phi_0)} \left(|r| e^{i\theta_r} + |tt'| e^{i(\theta_t + \theta'_t + \phi_2 - 2\phi_0)} |t| e^{i\theta_t} \right)$$

$$+ e^{i(2\phi_0)} \left(|r| e^{i\theta_r} + |tr'| e^{i(\theta_t + \theta'_r + \phi_2 + \phi_3 - 2\phi_0)} |t_{90}| e^{i\theta_{t_{90}}} |r| e^{i\theta_r} \right)$$

$$E_2 = -E_A \ (-) + E_{D_{90\%}} (+)$$

$$= e^{i(2\phi_0)} \left(|r| e^{i\theta_r} + |tt'| e^{i(\theta_t + \theta'_t + \phi_2 - 2\phi_0)} |r| e^{i\theta_t} \right)$$

$$+ e^{i(2\phi_0)} \left(|r| e^{i\theta_r} + |tr'| e^{i(\theta_t + \theta'_r + \phi_2 + \phi_3 - 2\phi_0)} |t_{90}| e^{i\theta_{t_{90}}} |t| e^{i\theta_r} \right)$$

Simplified,

$$I_1 = |E_1|^2$$

$$= \left| E_A \sqrt{0.5}\, e^{i\theta_t} + E_{D_{90\%}} \sqrt{0.5}\, e^{i\theta_r} \right|^2$$

$$= 0.5 |E_A|^2 + 0.5 |E_{D_{90\%}}|^2 + 2(0.5) |E_A| |E_{D_{90\%}}| \cos(\theta_t - \theta_r + \phi_A - \phi_{D_{90\%}})$$

$$I_2 = |E_2|^2$$

$$= \left| -E_A \sqrt{0.5}\, e^{i\theta_r} + E_{D_{90\%}} \sqrt{0.5}\, e^{i\theta_t} \right|^2$$

$$= 0.5 |E_A|^2 + 0.5 |E_{D_{90\%}}|^2 - 2(0.5) |E_A| |E_{D_{90\%}}| \cos(\theta_t - \theta_r + \phi_A - \phi_{D_{90\%}})$$

$$I_1 - I_2 = I_{detector}$$

$$\boxed{I_{detector} = (E_A\, E_{D_{aux}})^2 \cos(\theta_t - \theta_r + \phi_A - \phi_{D_{aux}})}$$

Drift in Laser power / coupling ratio of couplers :

For a Sudden power drift in Laser
or the Coupling ratio of the variable coupler. :

$P_{in} \rightarrow P_{in}'$ $\qquad$ $P_{in} \rightarrow$ E-field amplitude proportional $\sqrt{P_{in}}$

$E_{in} \rightarrow E_{in}'$ $\qquad$ $E_{in}' = \sqrt{P_{in}'}$

first variable coupler,

$E_{sam} \rightarrow E_{sam}'$

$E_{Ref} \rightarrow E_{Ref}'$

$E_{sam}' = \sqrt{k_1}\, E_{in}'$

$E_{Ref}' = \sqrt{1-k_1}\, E_{in}'$ $\qquad$ k_1 : Coupling ratio

Final variable coupler,

$$E_{out}' = \sqrt{k_2}\, E_{sam}'\, e^{i\phi_{sam}} + \sqrt{1-k_2}\, E_{Ref}'\, e^{i\phi_{Ref}}.$$ $\qquad$ k_2 : Coupling ratio

$$P_{out}' \propto |E_{out}'|^2 = \left|\sqrt{k_2}\sqrt{k_1}\, E_{in}'\, e^{i\phi_{sam}} + \sqrt{k_2(1-k_1)}\, E_{in}'\, e^{i\phi_{Ref}}\right|^2$$

$P_{in} \rightarrow P_{in}'$

$P_{out} \rightarrow P_{out}'$ $\Rightarrow$ $\dfrac{P_{in}'}{P_{in}}$

Now,

Sudden change in Coupling ratio of 1st variable coupler.

$E_{sam}' = \sqrt{k_1'}\, E_{in}$

$E_{Ref}' = \sqrt{1-k_1'}\, E_{in}$ $\qquad$ $k = 10 \log_{10}\left(\dfrac{P_{out}'}{P_{in}'}\right)$

at final coupler.

$$E_{out}' = \sqrt{k_2'}\, E_{sam}'\, e^{i\phi_{sam}} + \sqrt{1-k_2'}\, E_{Ref}'\, e^{i\phi_{Ref}}$$

then, $\quad \left|\sqrt{k_2'}\, e^{i\phi_{Sample}} \quad e^{i\phi_{Ref}}\right|^2$

then, $P_{out}' \propto \left| \sqrt{k_2}\sqrt{k_1'}\, E_{in}\, e^{i\phi_{Sample}} + \sqrt{k_2(1-k_1')}\, E_{in}\, e^{i\phi_{Ref}} \right|^2$

$\therefore$ Change in k_1 modifies the balance of amplitudes between two arms

this affects the interference contrast and asymmetrical fringes when $k_1' \neq 0.5$

- Sudden change of "k" at Final variable coupler

$$E_{out(A)}' = \sqrt{k_2'}\, E_{sam}'\, e^{i\phi_{sam}} + \sqrt{1-k_2'}\, E_{Ref}'\, e^{i\phi_{Ref}}$$

$$P_{out(A)}' \propto |E_{out(A)}'|^2$$

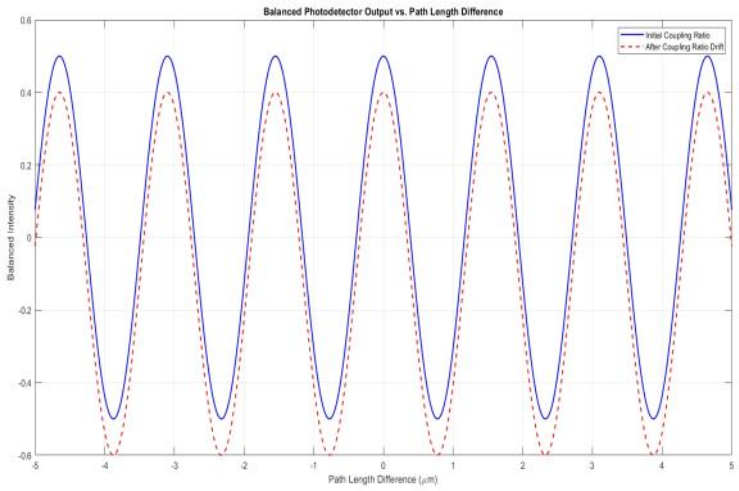
$$E_{out(B)}' = \sqrt{1-k_2'}\, E_{sam}'\, e^{i\phi_{sam}} - \sqrt{k_2'}\, E_{Ref}'\, e^{i\phi_{Ref}}$$

$$P_{out(B)}' \propto |E_{out(B)}'|^2$$

$\therefore$ $k_2' \neq 0.5$

no equal power distribution

Affects visibility



- Balanced Signal

→ Higher contrast

→ Low-freq noise Rejection

It gets rid of common-mode noise

→ Laser fluctuation

→ Coupling ration

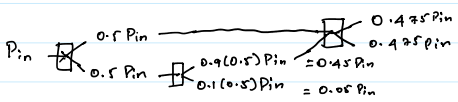
→ Environment noise

SNR ↑

advantage for locking.

$\cos(\Delta\phi(t))$ dependence

zero crossing peak trough detection



$= (0.5(0.5P_{in} + 0.45P_{in})) + (0.5(0.5P_{in} + 0.45P_{in}))$

$P = 0.475 P_{in}$

$= (0.5(0.5P_{in} + 0.45P_{in})) + (0.5(0.5\ P_{in} + 0.45P_{in}))$

$P_1 = 0.475\,P_{in}$

$P_2 = 0.475\,P_{in}$

$P_{in} = (0.475)P_{in} + (0.475)\,P_{in} + (0.05)\,P_{in}$

$$\boxed{P_{in} = P_{in}}$$